

Grounding, Hallucinations, and Fallback Answers

Grounded answers

A RAG answer is grounded when the language model bases its response on the passages retrieved from the document collection. The prompt should tell the model to use only the provided sources and avoid unsupported claims. Grounding does not guarantee perfect accuracy, but it helps limit answers to information that can be traced back to the local knowledge base.

Hallucinations

A hallucination occurs when a language model produces information that is unsupported, invented, or inconsistent with the available evidence. Retrieval can reduce hallucinations by supplying relevant source text before generation. However, the model can still make mistakes, so the application should keep the prompt strict and show the supporting source files to the user.

Fallback behavior

Sometimes the user asks a question that is not answered by any document in the knowledge base. In that case, the assistant should not guess. A safe fallback is a clear message such as: The information is not available in the provided sources. This behavior is especially important for a local assistant that is expected to answer only from a controlled document collection.

Why retrieval scores are not enough

Even when a question is unrelated to the knowledge base, a similarity search will still return the highest-ranked chunks unless a threshold or other rejection rule is used. Therefore, the generation step or a separate relevance check must decide whether the retrieved context actually contains enough information to answer the question.